

Usama Mirza

Ph.D. Student, Electrical and Electronics Engineering

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Research Interests: Generative models (diffusion models, GANs), computational imaging and inverse problems, accelerated MRI reconstruction, multi-contrast MRI synthesis, federated learning.

Education

- Ph.D. Electrical and Electronics Engineering**, Bilkent University, Ankara 2024–Present
Research: Diffusion-based methods for accelerated MRI reconstruction
Advisor: Prof. Tolga Çukur
- M.Sc. Electrical and Electronics Engineering**, Bilkent University, Ankara 2021–2024
CGPA: 3.77 / 4.00
Thesis: Diffusion Bridges for MRI Reconstruction
Advisor: Prof. Tolga Çukur
- B.Sc. Electrical Engineering**, SEECS, National University of Sciences and Technology (NUST), Islamabad, Pakistan 2017–2021
CGPA: 3.77 / 4.00
Final Year Project: Crop Monitoring using Computer Vision and IoT

Research Publications (Google Scholar)

Journal Publications

- M. U. Mirza**, O. Dalmaz, H. A. Bedel, G. Elmas, Y. Korkmaz, A. Güngör, S. Dar, K. K. Oğuz, and T. Çukur, “Learning Fourier-Constrained Diffusion Bridges for MRI Reconstruction,” *IEEE Transactions on Medical Imaging*, vol. 45, no. 6, pp. 3229–3245, June 2026.
- O. Dalmaz, **M. U. Mirza**, G. Elmas, M. Özbey, S. Dar, E. Ceyani, K. K. Oğuz, S. Avestimehr, and T. Çukur, “One Model to Unite Them All: Personalized Federated Learning of Multi-Contrast MRI Synthesis,” *Medical Image Analysis*, vol. 94, p. 103121, May 2024.

Conference Papers

- M. U. Mirza**, E. Ozdilek, E. Öztürk-Işık, and T. Çukur, “Low-Frequency Anchored Diffusion Transition Process for MRI Reconstruction,” in *Proc. 34th Signal Processing and Communications Applications Conference (SIU)*, Istanbul, Türkiye, July 2026.
- M. U. Mirza**, M. A. Aydın, and T. Çukur, “Frequency-Based Soft Diffusion Model for Accelerated MRI Reconstruction,” in *Proc. 33rd Signal Processing and Communications Applications Conference (SIU)*, Şile, Istanbul, Türkiye, June 2025.
- M. U. Mirza**, F. Arslan, and T. Çukur, “Super Resolution MRI via Upscaling Diffusion Bridges,” in *Proc. 32nd Signal Processing and Communications Applications Conference (SIU)*, Mersin, Türkiye, May 2024.
- O. Dalmaz, B. Sağlam, G. Elmas, **M. U. Mirza**, and T. Çukur, “Denoising Diffusion Adversarial Models for Unconditional Medical Image Generation,” in *Proc. 31st Signal Processing and Communications Applications Conference (SIU)*, Istanbul, Türkiye, July 2023.
- M. U. Mirza** and T. Çukur, “Super-Resolution Diffusion Model for Accelerated MRI Reconstruction,” in *Proc. 31st Signal Processing and Communications Applications Conference (SIU)*, Istanbul, Türkiye, July 2023.
- O. Dalmaz, **M. U. Mirza**, G. Elmas, M. Özbey, S. Dar, E. Ceyani, S. Avestimehr, and T. Çukur, “Personalized, Federated, and Unified MRI Contrast Synthesis,” in *Proc. IEEE 20th International Symposium on Biomedical Imaging (ISBI)*, Virtual, Apr. 2023.

- O. Dalmaz, **M. U. Mirza**, G. Elmas, M. Özbey, S. Dar, and T. Çukur, “A Specificity-Preserving Generative Model for Federated MRI Translation,” in *Proc. 3rd MICCAI Workshop on Distributed, Collaborative, and Federated Learning (DeCaF)*, LNCS vol. 13573, Virtual, Sept. 2022, pp. 79–88.
- **M. U. Mirza**, O. Dalmaz, and T. Çukur, “Skip Connections for Medical Image Synthesis with Generative Adversarial Networks,” in *Proc. 30th Signal Processing and Communications Applications Conference (SIU)*, Karabük, Türkiye, May 2022.

Conference Abstracts and Workshop Papers

- **M. U. Mirza**, O. Dalmaz, H. A. Bedel, G. Elmas, A. Güngör, and T. Çukur, “Accelerated MRI Reconstruction with Fourier-Constrained Diffusion Bridges,” in *Proc. 32nd Annual Meeting of the International Society for Magnetic Resonance in Medicine (ISMRM)*, Singapore, May 2024 (Oral; Summa Cum Laude Merit Award).
- **M. U. Mirza**, O. Dalmaz, H. A. Bedel, G. Elmas, A. Güngör, and T. Çukur, “MRI Reconstruction with Fourier-Constrained Diffusion Bridges,” in *Medical Imaging Meets NeurIPS Workshop*, New Orleans, LA, USA, Dec. 2023.
- O. Dalmaz, **M. U. Mirza**, G. Elmas, M. Özbey, S. Dar, E. Ceyani, S. Avestimehr, and T. Çukur, “A Personalized Federated Learning Approach for Multi-Contrast MRI Translation,” in *Proc. 31st Annual Meeting of the International Society for Magnetic Resonance in Medicine (ISMRM)*, Toronto, Canada, June 2023.
- O. Dalmaz, **M. U. Mirza**, G. Elmas, M. Özbey, S. Dar, E. Ceyani, S. Avestimehr, and T. Çukur, “pFLSynth: Personalized Federated Learning of Image Synthesis in Multi-Contrast MRI,” in *Medical Imaging Meets NeurIPS Workshop*, Virtual, Dec. 2022.

Academic Experience

- **Graduate Research Assistant**, ICON Lab, National Magnetic Resonance Research Center (UMRAM), Ankara 2021–Present
 - Developed Fourier-constrained diffusion bridges for accelerated MRI reconstruction (first author, IEEE TMI 2026; code: github.com/icon-lab/FDB).
 - Designed frequency-domain diffusion models for accelerated MRI and upscaling diffusion bridges for MRI super-resolution (first author, SIU 2023–2026).
 - Contributed to pFLSynth, a personalized federated learning model for multi-contrast MRI synthesis across institutions (Medical Image Analysis 2024).
- **Graduate Teaching Assistant**, Bilkent University, Ankara 2021–Present
Courses: Engineering Mathematics I & II, Linear System Theory, Neural Networks
- **Research Intern**, TUKL Research and Development Lab, Islamabad 2019
 - Surveyed FPGA accelerator architectures for deep neural networks.
 - Worked with Vivado HLS (C/C++) and Xilinx Vivado on Zynq FPGA platforms.
 - Prepared neural network models in Python for FPGA deployment.
- **Reviewer**, IEEE Transactions on Medical Imaging 2024–Present

Honors and Awards

- **ISMRM Summa Cum Laude Merit Award**, Singapore 2024
Oral presentation on Fourier-constrained diffusion bridges for accelerated MRI
- **Outstanding Cambridge Learner Award** 2015
Highest mark in the world in O-Level Mathematics.

Skills

- **Machine Learning**: Diffusion Models, Generative Adversarial Networks, Federated Learning
- **Frameworks & Libraries**: PyTorch, TensorFlow, Hugging Face, NumPy, SciPy, OpenCV
- **Programming Languages**: Python, C++, MATLAB
- **Tools**: Linux, CUDA (multi-GPU training), L^AT_EX, FFMpeg